Linear Programming: Part II

ORSA Study text

Faculty of Economics
and Management

Czech University
of Life Sciences

pef.czu.cz

Processed: Tereza Horáková

Sources: For this paper Chat GPT was used and Distance Learning Module for Management Science. Online. Available from: https://orms.pef.czu.cz/. [cit. 2024-10-30].

Position: Guarantor of ORSA, Lecturer
Workplace: Faculty of Economics and Management
Department of System Engineering

Czech University of Life Sciences Prague
Kamýcká 129, 165 00 Praha-Suchdol

Linear Programming

SENSITIVITY ANALYSIS

By examining the final (optimal) simplex tableau, we can learn a great deal. Besides showing the solution, the final tableau provides valuable information about the resources used, the range where the optimal decision remains unchanged, and the range where the coefficients in the objective function do not change the optimal solution. Specifically, it enables us to answer such questions as: Would you like to buy any more of a resource? If so, what price should you pay? How many units should you buy at that price? Similar questions can be answered about selling resources; even though a resource may be currently used in making products, at some price it is worthwhile to forgo production and sell it. These considerations are of interest because they lead to decisions that can increase profit or reduce cost.

Other questions are of the sort: If we change the profit per unit (by changing the coefficient in the objective function), will this change the optimal solution? This is sensitivity analysis, and refers to how much the solution changes for a small change in the objective function or, conversely, for a small change in the solution, the change that occurs in the objective function.

Other part of the sensitivity analysis deals with the impact of changes or available sources or other condition, which is expressed in the right sides of, constrains.

If there is a methodology for solving a decision model with one set of parameters, then the model can be solved again with a different set of parameters. The changes in the solution can then be evaluated based on the parameter change. This is the sensitivity analysis. In certain cases, it is more useful to calculate the amount that a parameter must change before the solution changes.

Sensitivity analysis involves:

- first finding the shadow price, the marginal value for a unit change in the right-hand side value of the constraint
- Second, finding the range of values over which the shadow price holds (the RHS ranges).

- Third, the objective function coefficient ranges. These ranges indicate the changes in the objective function coefficients within which the optimal solution remains the same.

In general, determining the sensitivity of the optimal solution to changes is easiest analytically when only one parameter changes.
For the sensitivity analysis we use the Example: Product planning with its initial and final tables:

		6	7	0	0	b	
		x_1	x_2	x_3	x_4		
0	x_3	2	3	1	0	24	
0	x_4	2	1	0	1	16	

Table 1: Initial tableau of the example Product Planning

		6	7	0	0	b	
		x_1	x_2	x_3	x_4		
7	x_2	0	1	1/2	-1/2	4	
6	x_1	1	0	-1/4	3/4	6	
	$c_j - z_j$	0	0	-2	-1	64	

Table 2: Optimal Solution of the example Product Planning

Economic Interpretation of the $c_j - z_j$ Values

First, consider the economic interpretation of the $c_j - z_j$ values, the bottom row of the simplex table. Recall that Z_j is the opportunity cost of introducing one unit of variable j into the solution-the cost of replacing or substituting for other solution variables. Since c_j is the unit profit, the $c_j - z_j$ value is the net profit (profit minus opportunity cost) resulting from introducing one unit of j into the solution.

In our example first consider the $c_j - z_j$ value associated with a slack variable, such as variable x_3 in Table 2. Variable x_3 is the slack variable associated with the first constraint, the limit on time available for machine 1. In the optimal solution in Table 2, $c_3 - z_3$ has a value of -2, meaning that introducing one unit of x_3 (i.e., one hour of slack time) into the solution will decrease profit by $2. Note that introducing one unit of slack into the solution is effectively the same as reducing the capacity of machine 1 by one hour. Hence, the $c_j - z_j$; values for slack variables associated with constraints can be considered as the marginal values associated with changes in capacity on that constraint. We called these marginal values dual prices. When a decision variable is not included in the optimal solution, the $c_j - z_j$ value represents the cost of forcing one unit into the solution.

Right-Hand Side Ranges

The interpretation of the dual price of x_3 in Table 2 is that one additional hour of machine time on machine 1 would be worth $2. We might ask the following question: If we could buy additional hours of machine 1 time for less than $2, could we increase our profit indefinitely by adding more hours?

We can answer this question by examining the x $_3$ column in Table 2. Recall the meaning of the coefficients in the table; if we were to introduce one unit of x_3, it would replace 1/2 unit of x_2, and 1/4 unit of x_1, (i.e., it would actually add 1/4 unit of x_1).

The question posed above, then, can be answered by considering how much x_3 can change before a change in the solution mix will occur.

If we enter positive x_3; into the solution, the interpretation is that additional slack hours are made available on machine 1-and thus, those hours to be used for production are reduced. In the table, the process of introducing x_3 is the same as doing an iteration in the simplex procedure with x_3 as the entering variable. We divide the "Solution values" column by the amount in the comparable row of the x_3 column (Table 3). The smallest nonnegative number thus obtained gives the number of units of x_3 :

		0	Solution values	
		x_3	B	Ω
7	x_2	1/2	4	8
6	x_1	-1/4	6	-24
	$c_j - z_j$	-2	64	

Table 3: Entering positive x_3

Here, the x_2 row is the only positive value and hence is also the smallest nonnegative value. This means that eight units of x_3 can be introduced before variable x_2 goes out of the solution. In terms of the problem, this means that we can cut back on a maximum of eight hours of those available on machine 1 before production of A product stops.

We can also consider adding negative x_3. This means that we make additional hours available on machine 1. In order to consider introducing negative x_3 into our table, we can proceed as above, except that we must multiply all the values in the x_3 column by (- 1) before going to the next step in the simplex procedure. Thus, we would have Table 4:

		0	Solution values	
		x_3	B	Ω
7	x_2	-1/2	4	-8
6	x_1	1/4	6	24
	$c_j - z_j$	-2	64	

Table 4: Entering negative x_3

As before, the smallest nonnegative value determines how many units of negative x_3 can be introduced. Here, it is 24; after adding 24 additional hours of time (i.e., negative x_3) on machine 1, variable x_1, goes to zero and out of the solution.

The two steps can be combined into one. Look at the ratios of the "Solution values" column to the slack variable column. The smallest nonnegative ratio determines the amount of decrease; the smallest (in the sense of being closest to zero) nonpositive ratio determines the amount of the increase. For purposes of this rule, a ratio of zero divided by a positive number is considered positive, and zero divided by a negative number is negative.

We have derived a range of values relative to the amount of time available on machine 1. We started out with 24 hours available. When, we found that we could take away 8 hours before the solution mix changed, or we could add 24 hours before the solution mix changed. Hence, we have a range of 16 to 48 for hours available on machine 1, over which the basic solution mix does not change. Recall that the dual price for a marginal hour of machine 1 time was $2 (the $c_3 - z_3$ value). This value holds over the range we have just determined. In short, machine time on machine I has a value of $2 per hour over a range of available hours from 16 to 48 (assuming machine 2 hours remain fixed at 16).

Note that 16 to 48 hours is a range within which the solution mix remains the same. That is, the list of solution variables does not change. The actual solution values will change as more or less time is available on machine 1. Also, total profit will change at a rate of $2 per hour (the marginal value for x_3) for increments of machine 1 time.

We can do the same type of analysis on the time available on machine 2. Variable x_4 is the slack variable associated with machine 2. Look at the ratio of the "Solution Values" column to the slack variable column for x_4:

		0	Solution values	
		x_4	B	Ω
7	x_2	-1/2	4	-8
6	x_1	3/4	6	8
	$c_j - z_j$	-2	64	

Table 5: Entering x_4

As indicated above, the smallest nonnegative value determines the amount of decrease; the smallest (in the sense of being closest to zero) nonpositive ratio determines the increase. Machine 2 has 16 hours available and both the increase and decrease in this case is 8, or a range from 8 to 24 hours.

A summary of these results is shown in Table 6.

Resource	Machine hours Available	Dual price	Range over which dual price is valid	
	(original RHS)		Lower Limit	Upper Limit
Machine 1	24	2	16	48
Machine 2	16	1	8	24

Table 6: RHS Ranges

In some cases, all of the available capacity of a constraint is not used, and the slack variable will have a positive value in the solution. The dual price in such cases is zero-all of the available capacity is not being used, so additional capacity has no value. In that case, there is no upper limit on the range within which the zero dual price holds. But how much could the capacity be reduced before the solution changed? The answer is simple-if there are, for example, eight hours of slack in a given constraint, then capacity can be reduced by this eight hours before the solution is forced to change.

Changes in the Prices

Another major concern to managers is the sensitivity of the linear programming solution to changes in prices: that is, to changes in the per-unit profits (or costs) of variables in the objective function.

In the case of variables that are not in the solution, determining this sensitivity is relatively easy. Recall that the Z_j measures the opportunity cost of introducing one unit of the particular variable into the solution. If the variable is not in the optimum solution, it means that its profitability (c_j) is not as great as the opportunity cost Z_j . In other words, $c_j - z_j$ is negative (This is true if we are maximising. If we are minimising, then the variable will come into the solution if the cost drops below Z_j.) To come into the solution, the profit must exceed Z_j. This is the upper limit before a change occurs. On the other hand, since the variable is not currently profitable enough to be in the solution, any decrease in per unit profit will not change its status.

In summary, for a variable not in the optimum solution, the price ranges are:

Lower Limit: no

Current Value: c_j

Upper Limit: Z_j

Within these limits, there is no change in the optimal solution.

For variables that are already in the solution, determining the sensitivity to per-unit profit changes is not so simple. The dual to a linear programming problem is a complementary problem, the details of which will be explained. The same kind of analysis used to determine the right-hand side ranges can be used on the dual problem to determine the profit coefficient ranges.

Consider variable x_1, which had an original price of \$6 and is a solution variable. We divide the coefficients of the $c_j - z_j$ row in Table 2 by the coefficients of the x_1, row This is the dual problem equivalent of dividing the "Solution values" column by the slack variable column used for right-hand side ranging. Then (see Table 7), the smallest nonnegative ratio obtained determines the amount of the increase in price, and the smallest nonpositive (closest to zero) ratio determines the decrease in price (exclude thex_1, column from these calculations).

		7	0	0
		x_2	x_3	x_4
6	x_1	0	-1/4	3/4
	$c_j - z_j$	0	-2	-1
$(c_j - z_j)$ row / x_1 row		/	8	-4/3

Table 7: Possible changes in c_1 Price

Thus, the possible profit increase is \$8 and the decrease is \$4/3 for variable x_1. That is, the per-unit profit can range from \$6 - \$ 1.33 = \$4.67 to \$6 + \$8 = \$14 without any change in the solution.

A similar analysis can be done for variable x_2, which price can vary from \$3 to \$9. Recall that the solution of this problem called for the production of six units of x_1 and four units of x_2. That this solution will not change unless the profit from a unit of x_1, drops below \$4.67 per unit, or increases to above \$14 per unit. Similarly, the solution will not change unless the x_2 profit per unit falls below \$3 or increases above \$9. Within the ranges specified by the objective function values, both the solution mix and the solution values remain the same. Total profit changes, however, when the unit profits (individual values) change.

The kind of information contained in sensitivity analysis is very valuable to management. The prices are usually subject to change from time to time. If a change is within the range determined by the sensitivity analysis, then there is

no effect upon the optimum solution. If the change is outside the range, a new solution is needed, and the programming problem must be re-solved.

The linear programming problem is set up as a decision problem under certainty. Few problems in the real world are truly certain. Often, many factors are unknown and must be estimated using the best judgement available. The use of sensitivity analysis helps to show over what ranges the solution is and is not subject to change. As such, it is an important adjunct to the interpretation of the solution of a linear programming problem.

Summary
Detail examination of the optimal results including sensitivity analysis is at least as important as the solution itself. The final solution gives many answers useful for manager's decision making and represents the real power and value of LP.

Shadow prices can be useful in identifying bottlenecks or constraints that are cost and might be profitably changed. Shadow prices can also be useful in evaluating products.

The ranges, both for the RHS coefficients and for the objective function coefficients, provide important information in interpreting the LP solution. The RHS ranges determine the limits within which the shadow price holds for each constraint. The objective coefficient ranges determine limits within which the solution remains the same.

The sensitivity analysis should be included in software product used for solving LP models. The user must first know it the analysis is really available and that he or she must learn how to use and interpret the computer output.

THE DUAL PROBLEM

Every linear programming problem we have solved has been of the type designated as primal, the primal being the first problem to which our attention is generally directed. Each primal problem has a companion problem called the dual. The dual has the same optimum solution as the primal, but it is derived by an alternative procedure. The analysis of this procedure may be instructive for several types of decision problems.

Frequently, the economic problem being solved as the primal involves the maximisation of an objective function (for example, profits), subject to constraints that are frequently of a physical nature (such as hours of machine time). In this type of situation, the dual involves the minimisation of total opportunity costs, subject to the opportunity cost (or equivalently, the value) of the inputs of each product being equal to or greater than the unit profit of the product.

There are five steps to obtain the dual from the primal problem:

1. If the objective function is maximised in the primal, the objective function of the dual is minimised. If the objective function of the primal is a profit equation, the objective function of the dual will be a cost equation.

2. The coefficients of the variables of the cost equation (the dual objective function) are the right-hand side constants of the primal constraints.

The constants for the dual constraints are obtained from the profit function (the objective function) of the primal.

4. Transposing the coefficients used in the primal forms the dual constraints. The first column of A becomes the first row of A^T, and the second column of A becomes the second row of A^T.

5. If we are maximising a primal objective function and if the constraints of the primal are "less than or equal to," the constraints of the dual will be "greater than or equal to."

If this entire procedure is performed on the dual, the original (primal) problem will be obtained. Thus, the dual of the dual is the primal.

Rules for Constructing Dual Problems

These rules require the primal problem to be either a maximisation problem with all "less than or equal to" constraints, or a minimisation problem with all "greater than or equal to" constraints. If an LP problem does not have one of these forms, multiplying the objective function or the constraints by (- 1) and writing equality constraints as two inequality constraints will enable it to be placed in one of the forms required by the rules listed above.

Dual Solution

The economical interpretation of the dual solution is of considerable importance and usefulness.

An important fact is that the dual of dual is the original model. The values of the dual variables of the dual problem are the optimal values of the original model's variables. I f the original model has m constraints and n variables, its dual has m variables and a constraints. If the number of constraints is much greater than the number of Variables, solving the dual problem is easier. Unless the computational saving are significant, you should solve the model that interests you, not its dual. Sensitivity analysis and model modification are easier on the original model.

Summary

Although the primal and dual problems use the same numerical values, the relationship between the two is not obvious. If one model is ill-behaved, then so is the other. If one model has a finite optimal solution, then so does the others and the optimal objective function values are equal. The are relationship between constraints in one model and variable in the other. The interpretation of the dual constraints should help clarify the relationship between the primal and dual problems. Solving the dual problem brings answers to another questions than the primal solution. The dual theory is a base for sensitivity analysis of the prices.